

Month 1: Purple Teaming Batch 01

Network Security & Protocol Analysis

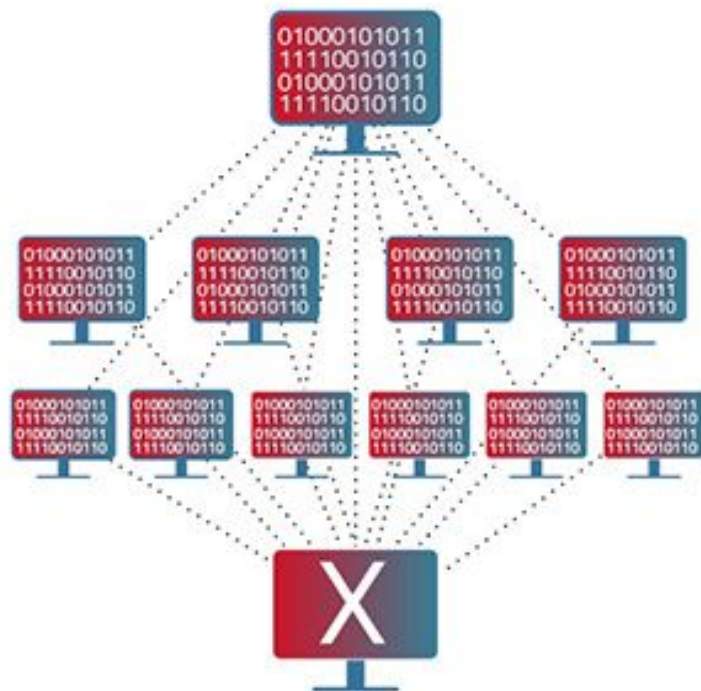
Week 3: Network Protocol Abuse

Instructor: **Muhammad Saad**

01000101011
11110010110
01000101011
11110010110



DoS attack



DDoS attack



Threat Actor

Malicious packet-larger then 110,000 bytes

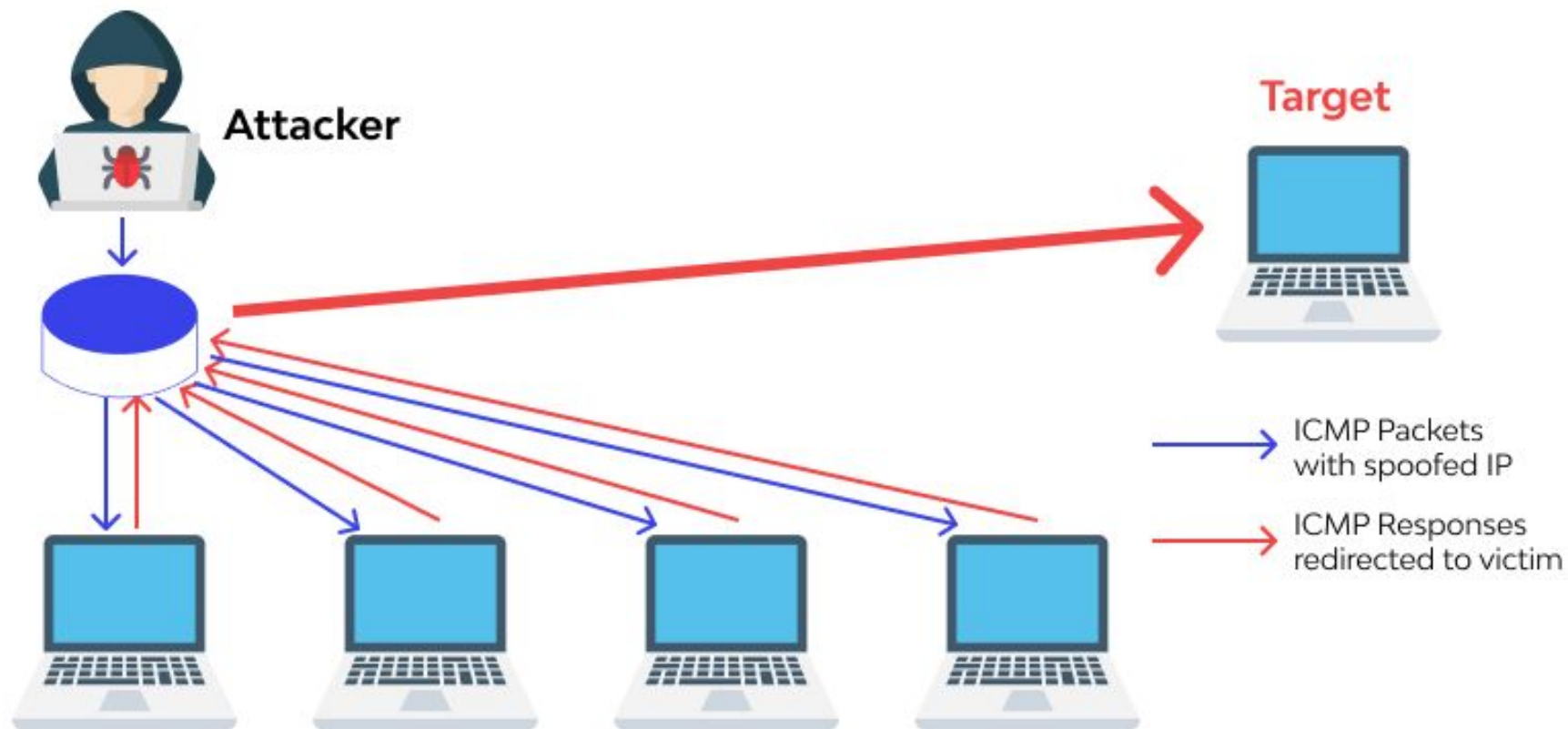


Normal IP packet-maximum size: 65,538 bytes



Victim

Smurf Attack



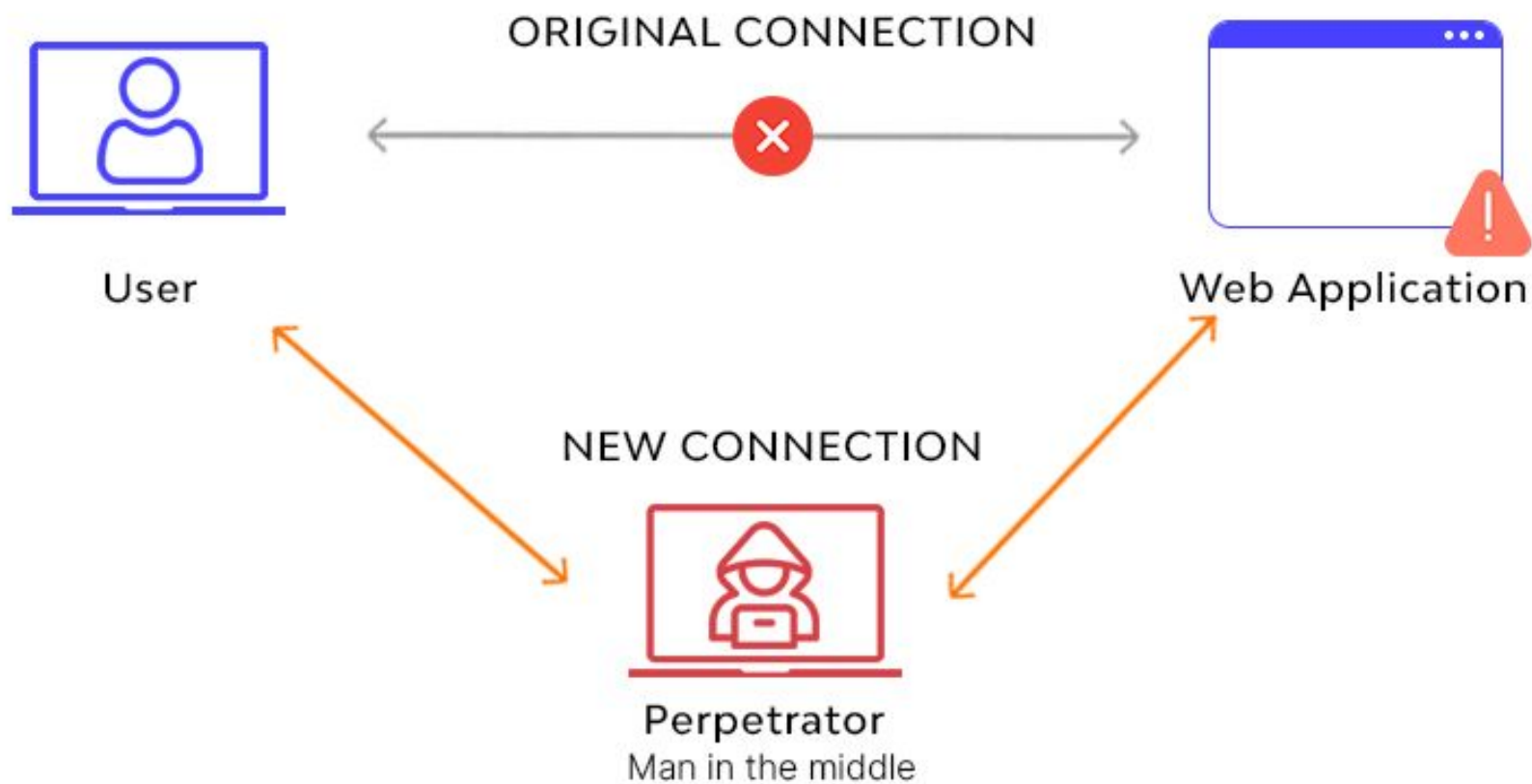
MITM – Man-in-the-Middle

- A **Man-in-the-Middle (MITM) attack** is a **cyberattack** where an **attacker** **secretly intercepts** and **possibly alters** **communication between two parties** who **believe they are communicating directly with each other**. It is a concept in the field of Cybersecurity.
- **How it works:**
 - **Two users start communication** (for example, your computer and a website).
 - **The attacker places themselves in between** the connection.
 - The attacker can **read, steal, or modify the data** being transmitted.

So instead of: **User → Website**

it becomes: **User → Attacker → Website**

Both the user and the website think they are talking directly to each other, but the attacker is controlling the communication.

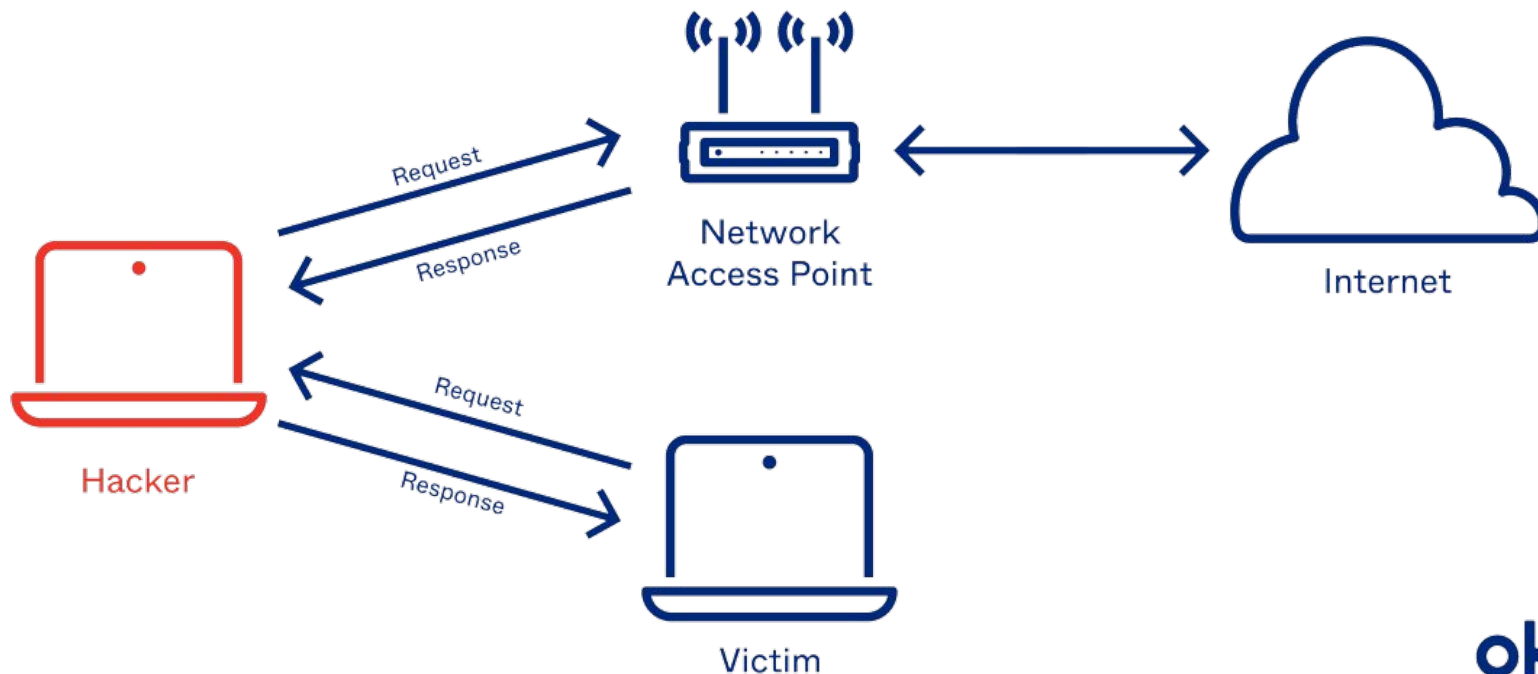


ARP Spoofing – ARP Poisoning

- **ARP Spoofing** (also called **ARP Poisoning**) is a type of **network attack** where an attacker sends fake **ARP messages** on a local network to associate their MAC address with the IP address of another device (like the router). This attack is commonly used to perform a Man-in-the-Middle attack.
- **How ARP works:**
 - Device wants to send data to **192.168.1.1**
 - It asks: *“Who has this IP?”*
 - The router replies with its **MAC address**
- **How ARP Spoofing Works:**
 - Attacker joins the same network (e.g., public Wi-Fi).
 - The attacker sends **fake ARP replies** to devices.
 - Devices update their ARP table with the attacker’s MAC address.
 - Traffic meant for the router or another device goes **through the attacker first**.

Victim → Attacker → Router

ARP Poisoning/Spoofing



Best of Luck for the Week 3 Tasks

Instructor: **Muhammad Saad**